



Reverse query $\leftrightarrow$ swap target

$q_+ : \text{dog followed by rooster} \rightarrow W_+$

$q_- : \text{rooster followed by dog} \rightarrow W_-$

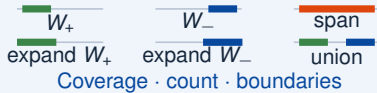
(b) RelTwin training

Stage 1: Answer learning

	$y(W_+)$	$y(W_-)$
q_+	$s_{++} \uparrow$	$s_{+-} \downarrow$
q_-	$s_{-+} \downarrow$	$s_{--} \uparrow$

Competition + reversal consistency

Stage 2: Set preferences



Inference

Audio + query $\longrightarrow$ Adapted backbone $\longrightarrow$ Timestamps

(c) Window selection

Paired SFT

0 27 s

IoU 0.00

q_+

IoU 0.98

q_-

target

+ Candidate

0 27 s

IoU 0.69

q_+

IoU 0.98

q_-

prediction